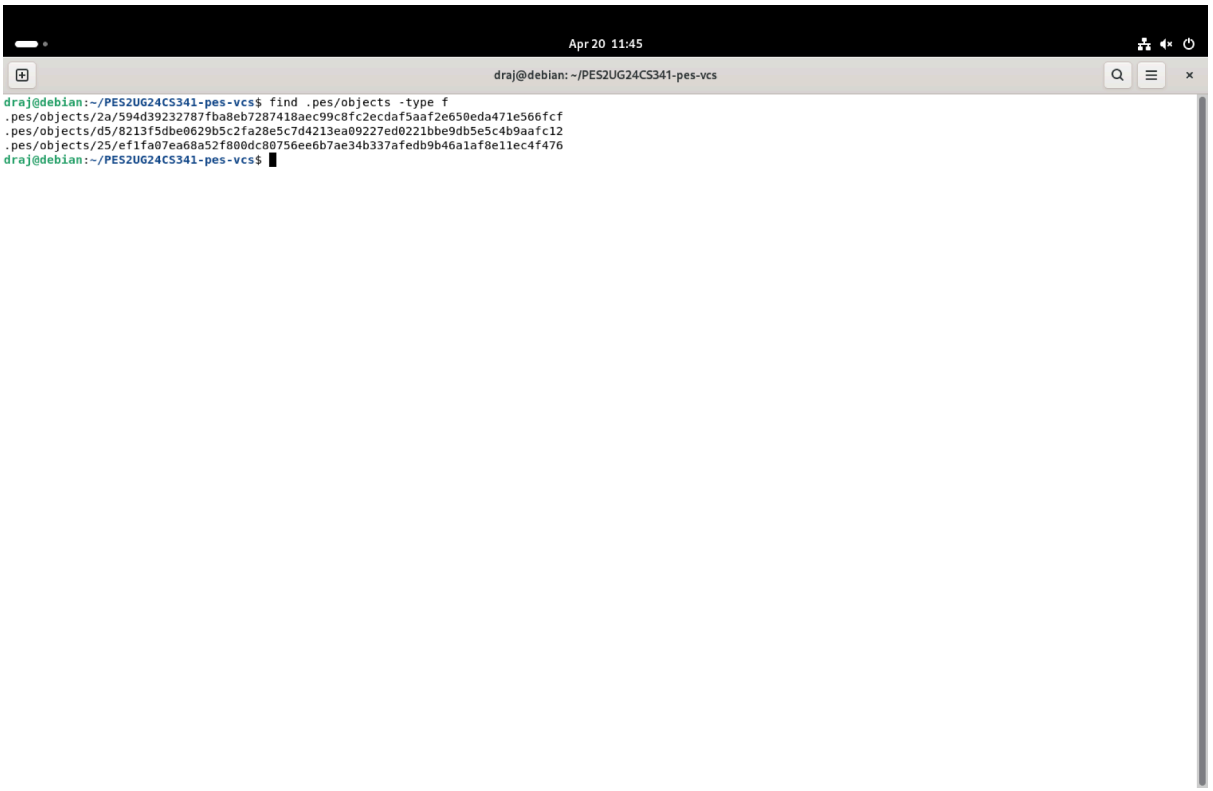
1A)

```
draj@debian:~/OS-U4-ORANGE$ make test_objects
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
draj@debian:~/OS-U4-ORANGE$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
draj@debian:~/OS-U4-ORANGE$
```

1B)



PHASE 2

2A)

```
draj@debian:~/PES2UG24CS341-pes-vcs$ make test_tree
./test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
draj@debian:~/PES2UG24CS341-pes-vcs$
```

2B)

```
draj@debian:~/PES2UG24CS341-pes-vcs$ find .pes/objects -type f
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
draj@debian:~/PES2UG24CS341-pes-vcs$
```

```
draj@debian:~/PES2UG24CS341-pes-vcs$ xxd .pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f | head -20
00000000: 626c 6f62 2036 0077 6f72 6c64 0a        blob 6.world.
draj@debian:~/PES2UG24CS341-pes-vcs$
```

PHASE 3

3A)

```
draj@debian:~/PES2UG24CS341-pes-vcs$ ./pes init
./pes add file1.txt file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  object.c
  untracked:  index.h
  untracked:  Makefile
  untracked:  index.c
  untracked:  .gitignore
  untracked:  test_sequence.sh
  untracked:  commit.c
  untracked:  tree.c
  untracked:  test_objects.c
  untracked:  pes.c
  untracked:  README.md
  untracked:  test_tree.c
  untracked:  commit.h
  untracked:  tree.h
  untracked:  pes.h

draj@debian:~/PES2UG24CS341-pes-vcs$
```

3B)

```
draj@debian:~/PES2UG24CS341-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776667456 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776667456 6 file2.txt
draj@debian:~/PES2UG24CS341-pes-vcs$
```

PHASE 4

4A)

```
draj@debian:~/PES2UG24CS341-pes-vcs$ ./pes commit -m "message"
Committed: f40ad1704c3e... message
draj@debian:~/PES2UG24CS341-pes-vcs$
```

4B)

```
draj@debian:~/PES2UG24CS341-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/f4/0ad1704c3ee9e9fe8d08ec93de86d297bb385fc31a10bab3a5a3b12f557de5
.pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2
.pes/refs/heads/main
draj@debian:~/PES2UG24CS341-pes-vcs$
```

4C)

```
draj@debian:~/PES2UG24CS341-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
f40ad1704c3ee9e9fe8d08ec93de86d297bb385fc31a10bab3a5a3b12f557de5
ref: refs/heads/main
draj@debian:~/PES2UG24CS341-pes-vcs$
```

FINAL INTEGRATION

```
draj@debian:~/PES2UG24CS341-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: e9001950b054... Initial commit

Log after first commit:
commit e9001950b05421a6bcbd1b94f1b17ee857b61abbd8bc16aeab5bc79662c47e03
Author: Dheeraj <PES2UG24CS341>
Date:   1776671452

    Initial commit


--- Second Commit ---
Committed: b93f7c6fda24... Update file.txt

--- Third Commit ---
Committed: e0ce87f349f5... Add farewell

--- Full History ---
commit e0ce87f349f5ca048f8ed01feaa95b5193b2af0c01d68a50e4328117f7f4a39d
Author: Dheeraj <PES2UG24CS341>
Date:   1776671452

    Add farewell

commit b93f7c6fda24db4aaa75634a41f6c00a927555dea88429e16be0e9ea11d0bb62
Author: Dheeraj <PES2UG24CS341>
Date:   1776671452

    Update file.txt

commit e9001950b05421a6bcbd1b94f1b17ee857b61abbd8bc16aeab5bc79662c47e03
Author: Dheeraj <PES2UG24CS341>
Date:   1776671452

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
e0ce87f349f5ca048f8ed01feaa95b5193b2af0c01d68a50e4328117f7f4a39d

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/b9/3f7c6fda24db4aaa75634a41f6c00a927555dea88429e16be0e9ea11d0bb62
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e0/ce87f349f5ca048f8ed01feaa95b5193b2af0c01d68a50e4328117f7f4a39d
```